

GW — lab notebook (gravitational-wave program)

K. Bhatia ORCID 0009-0007-4447-6325 repo: GW (ported from kunalb541/cosmo2, E-numbering preserved)
Data of record: GWTC-2.1/3 PE (GWOSC/Zenodo, 76 events); GWTC-4.0 PE (Zenodo 16053484, 86 events);
LVK TGR products (Zenodo 17461225: imr/par/rin/liv); GWTC-3 cosmology icarogw (Zenodo 5645777); PTA
chains: NANOGrav 15yr fig-1 release, EPTA DR2 (Zenodo 8091568), PPTA DR3 (github). Every battery locked-
preregistered before running; every headline independently re-derived; numbers of record in **results/**, per-battery
detail in **reports/**.

Black-hole posterior geometry: the event atlas and its laws (E38, E40–E44)

E38 GW posterior-geometry atlas. Value/shape geometry on real compact-binary posteriors: all 76 confident GWTC-1/2.1/3 events (1.54M samples; GWOSC/Zenodo PE releases), 6 black-hole planes each — (m_1, m_2) , $(\mathcal{M}, q)$, (q, χ_{eff}) , (M_f, a_f) , $(d_L, \cos \iota)$, (z, M_{tot}) ; 74/76 produce all six. The foundation for E40–E47.

E40 GW chirp-mass lawfulness. Is the (m_1, m_2) posterior degeneracy just the constant-chirp-mass line? Predict each event's orientation from the constant- $\mathcal{M}$ tangent at its median masses, vs a constant-total-mass null, 75 events. PASS: median $|\Delta\psi_{\text{chirp}}| = 7.49^\circ$ vs total-mass 20.38° ; chirp wins 61/75; on the elongated subset (axis ratio ≥ 3 , $n=28$) 4.68° ; Spearman(error, axis-ratio) = -0.42 (more elongated $\rightarrow$ tighter, as predicted). A rare POSITIVE geometric-lawfulness result — geometry is lawful where a dominant physical combination (chirp mass) exists.

E41 GW plane lawfulness map. A-priori correlation-sign predictions for all 6 planes, derived by 12 physicist/verifier agents blind to the data, tested against measured within-event ρ (75 events). 3/6 HIT:

E42 BH population + precision lawfulness. Precision test: chirp-mass fractional precision $\sigma_{\mathcal{M}}/\mathcal{M}$ vs SNR (75 events). PASS: $r = -0.50$ ($p=7\times 10^{-7}$), OLS slope -1.22 (Fisher -1), and partial correlation with $\log \mathcal{M}$ at fixed SNR = $+0.94$ (the cycle-count law $N \propto \mathcal{M}^{-5/3}$). Posterior SIZE is lawful — completing orientation (E40) + correlation (E41) + size (E42). Population map: χ_{eff} mean $+0.076$, 36% median-negative, m_1 peaks near 10 and $35 M_\odot$, 7 IMBH-scale remnants, 1 BNS; the geometric outliers are the physically extreme events (GW190814 $q=0.11$, GW191219 $q=0.04$ NSBH, GW200115 NSBH, GW190425 BNS).

E43 χ_{eff} per-event negatives. Does any event securely have $\chi_{\text{eff}} < 0$ (dynamical-formation signature)? From the full posteriors (75 events): 0 events with $P(\chi_{\text{eff}} < 0) > 0.90$ (GW191109 sits at exactly 0.900, 90% CI $[-0.60, +0.12]$ spans 0), 0 with a 95% CI below 0. NULL: E42's 36% median-negative is largely measurement scatter, not per-event-secure negatives.

E44 χ_{eff} hierarchical. Does the POPULATION require support below 0? Gaussian fit (flat and induced-prior-reweighted, 75 events) + a positive-truncation comparison. $\mu \approx 0.06\text{--}0.09$, $\sigma \approx 0.045\text{--}0.061$, negative fraction median $\sim 9\%$ — but a $\chi_{\text{eff}} \geq 0$ truncated population fits as well or better ($\Delta \ln L = -1.0/-0.74$), so the negative tail is a Gaussian-tail artifact, NOT required. Reinforces E43. (Does not overturn the LVK negative tail: a simplified pipeline — Gaussian-only, approximate prior, no selection function — simply does not independently demand it.)

Testing general relativity with black holes (E45–E47, E55, E57–E60)

E45 IMR-consistency test. LVK GWTC-3 Tests-of-GR: the inspiral- vs post-inspiral-inferred final mass/spin, deviation $(\Delta M_f/M_f, \Delta a_f/a_f)$, GR = (0, 0), 18 events. GR UPHELD — D_1 combined deviation $(-0.035, +0.002)$, GR at credible level 0.671; D_2 17/18 consistent (outlier GW190814 has an absent post-inspiral, a known systematic). D_3 (the framework's added test): the mean-deviation directions are COHERENT (Rayleigh $p = 0.001$, 13/18 with positive final-spin deviation), only partly along the shared degeneracy — a coherent sub-threshold waveform systematic that per-event tests miss.

E46 parameterized PN deviations. Fractional PN inspiral-phase deviations $\delta\hat{\varphi}_i$ (GR = 0), 9 events $\times$ 10 coefficients (SEOBNRv4). GR NOT violated. The cross-event sign-coherence test isolates **dchi0** (leading 0PN):

9/9 events positive, Bonferroni $p = 0.004$, combined $+0.109 [0.040, 0.177]$ excluding 0 — the documented low-PN WAVEFORM SYSTEMATIC of a single-waveform analysis, not new physics. The non-coherent combined exclusions (`dchi3/51/61`, only 2–4/9 positive) are naive-product artifacts the coherence test correctly discounts. Single waveform; multi-waveform robustness (not run) would confirm the systematic origin.

E47 ringdown no-hair test. Black-hole spectroscopy: fractional deviation of the 221 overtone frequency/damping ($\delta f, \delta \tau$) from the Kerr prediction, GR = (0, 0), 22 events. GR/Kerr UPHELD — D_1 combined 0.75σ from GR ($Q = 0.246$, via a robust Gaussian product); D_2 22/22 consistent; D_3 ISOTROPIC (Rayleigh $p = 0.79$), no coherent deviation (unlike E45/E46 — the coherent systematics are inspiral-specific, absent in the overtone). A first-pass histogram-density product spuriously reported a 99.9% “no-hair violation” that contradicted D_2/D_3 (an event at $Q=0.987$ was only 0.25σ from GR); the Mahalanobis cross-check caught the binning artifact and the Gaussian product corrects it.

E55 multi-waveform robustness. E40’s chirp-mass orientation across two independent waveform families (IMRPhenomXPHM vs SEOBNRv4PHM, 59 events): PASS — the law holds for both (median $|\Delta\psi_{\text{chirp}}| 9.1^\circ/6.1^\circ$) and the orientations agree to 3.3° (elongated). Real geometry is waveform-robust; E46’s low-PN `dchi0` coherence, by contrast, is a known waveform systematic (would not survive). DATA-BLOCK (honest): a direct same-event multi-waveform test of the E45/E46 GR-test coherences is impossible with GWTC-3 (the parameterized set is FTI/SEOBNR-only; TIGER/IMRPhenom was delayed) — would need the GWTC-2 (different events).

E57 black-hole area theorem. Hawking’s 2nd law from GWTC-3: $A_{\text{final}} \geq A_{\text{initial}}$ using the inspiral progenitors (initial areas) vs the ringdown remnant (final area), disjoint/ independent (Isi 2021 setup). Only 4 loud events carry a ringdown remnant. HELD: no violations, area increases 5–79%, GW170104 strong single-event confirmation 0.92, stacked 0.92σ . Honest: crude independent-pairing + broad inspiral spin priors give GW150914 $P=0.74$, below Isi’s 0.97 — a consistency check, not a stringent bound.

E58 massive graviton / Lorentz-violation. GW-propagation gravity test: modified dispersion $E^2 = p^2 c^2 + A p^\alpha$ over 12 events $\times$ 8 orders ($\alpha=0$ = graviton mass). GR UPHELD: no Lorentz-violation/dispersion signal in any event or order. Graviton mass (per-event, robust): best single-event $m_g < 6.7 \times 10^{-24}$ eV/ c^2 , median 1.8×10^{-23} (LVK combined 1.27×10^{-23}). Honesty catch: a naive posterior-product combined bound gave an absurd 5.9×10^{-27} (3 orders too tight, grid-unstable) — LVK use a flat-in- A prior and the prior samples are empty in the release, so it cannot be divided out; the per-event bounds are reported and the LVK combined quoted, not a fabricated number.

E59 ringdown mode consistency. Does adding the 221 overtone shift the remnant (M_f, a_f) from the 220-fundamental value? `rin.zip` Kerr_220 vs plain Kerr_221, 22 events. A naive test “fails” (7 events $> 2\sigma$, max 6.1), BUT the shift is COHERENT — M_f decreases in 22/22 (100%) and a_f in 21/22 (95%) with the overtone. A uniform all-event down-shift is the known ringdown- overtone SYSTEMATIC (start-time/modeling; the GW150914 overtone controversy), NOT a no-hair violation. The coherence lens (as in E45/E46) prevents a false violation claim; the clean no-hair result remains E47 (221 deviation params ~ 0 , isotropic).

E60 model-independent ringdown no-hair. The strongest-in-principle ringdown test: the MEASURED 220 QNM frequency (damped-sinusoid fit, no Kerr assumption) vs the Kerr prediction $f_{220}(M_f, a_f)$ from the INSPIRAL remnant (Berti fit) — independent data. Kerr CONSISTENT: no event deviates $> 2\sigma$ (max $|z|=0.85$), combined 0.34σ . GW150914 clean: Kerr 236.6 vs measured 246.8 Hz (+5%). Weak (only 3 events have both a DS ringdown and an inspiral remnant; GW170814’s DS frequency is broad/poorly constrained), GW150914 carries the confirmation. Ringdown no-hair trilogy: E47 (deviation params ~ 0) + E59 (remnant shift = systematic) + E60 (frequency matches Kerr) all uphold the Kerr nature.

Deriving the mass-plane law; the curved chirp-mass law (E65, E67)

E65 the E40 law derived — locked Fisher-rotation REFUTED; the posterior IS the constant- $\mathcal{M}_c$ curve. Locked hypothesis: E40’s residual pattern (heavy events fail toward constant- M_{tot}) is a Fisher-mixture rotation — inspiral cycles measure $\mathcal{M}_c$, the merger band measures M_{tot} , so orientation should rotate with detector-frame mass. Model: two-direction Fisher mixture, analytic aLIGO PSD, one constant κ fit ONLY on the 14 locked E40-prereg events, scored out-of-sample on the other 61. OUTCOME: D1 marginal pass (7.45 vs

7.72°), D2 FAIL, D3 (parameter-free sign test) FAIL decisively — measured ψ does NOT sit between the chirp and total tangents; rotation fractions are mostly NEGATIVE (light elongated events overshoot PAST the chirp tangent, away from total), a sign no rotation model can produce. What the failure revealed (post-hoc, labelled): the posterior lies along the constant- $\mathcal{M}_c$ CURVE; the principal axis of a curved arc is its chord, not the median-point tangent. Zero-free-parameter curved law (uses only the $\mathcal{M}_c$ median + q marginal): median error 4.85 vs 7.49° (all 75), **0.81** vs 4.68° (elongated, 6×); GW190728’s 14° “violation” reproduced to 1°. On-record GWTC-4 out-of-sample prediction: $< 2^\circ$ for elongated O4a events. 5 contract tests; transient ernno-60 read handled by retry; headlines re-derived by hand.

E67 GWTC-4 out-of-sample: the curved chirp-mass law PASSES at 1.26°. The E65 prediction — locked on record (commit 8d09cbf) from GWTC-1/2.1/3 data only, prereg pushed BEFORE any GWTC-4 file was opened — scored on the fully disjoint GWTC-4.0 O4a catalog (Zenodo 16053484; 86 events, all processed, zero dropped, all files h5py-validated). **D1 PASS: median $|\psi_{\text{curve}} - \psi_{\text{meas}}| = 1.26^\circ$ on the 19 elongated events (threshold 2°)** — 18 of 19 land at 0.2–3.3° while the tangent carries systematic 4–11° errors that the curvature term removes (GW231118_090602: 10.7 $\rightarrow$ 0.5; GW231020: 8.0 $\rightarrow$ 0.2; GW230529 mass-gap: 2.9 $\rightarrow$ 0.3). D2 PASS: curve beats tangent on all 86 (7.53 vs 9.36°). D3: the E40 tangent law itself replicates out-of-sample (86% wins vs 81%; Spearman -0.40 vs -0.42). The E65 finding is hereby PROMOTED from post-hoc to out-of-sample confirmed: the (m_1, m_2) posterior geometry of GW events is the constant- $\mathcal{M}_c$ curve (zero free parameters — each prediction uses only the event’s own median source $\mathcal{M}_c$ and q marginal). Headlines re-derived independently + one event recomputed end-to-end by hand (exact match).

Gravitational-wave cosmology: the spectral-siren lever (E66)

E66 spectral-siren H_0 obstruction — REFUTED at GWTC-3 precision. Radical claim (E61’s obstruction logic on GW): the spectral-siren H_0 program is blocked by the $\sim 33 M_\odot$ mass-peak location μ_g acting as an uncalibrated ruler. Test: linear-response couplings on the REAL LVK GWTC-3 joint cosmology+population posterior (icarogw release, $n = 13,533$; power-law+ peak, restricted flat- Λ CDM). Locked stop-and-diagnose fired mid-run: the multivariate response $C_{AB}C_{BB}^{-1}$ (partial) disagreed with the mandatory decile cross-check (marginal) — an apples-to-oranges comparison in the implementation, caught exactly as the two-estimator rule intends; fixed to the marginal degeneracy track, after which all sanity gates pass. RESULT: a $4 M_\odot$ coherent μ_g drift moves H_0 by only -0.51 km/s/Mpc ($< 1 \Rightarrow$ REFUTED; robust $-0.41/-0.32$ across SNR cuts) — an order of magnitude below the CMB-vs-SH0ES gap and inside the chain’s own ± 3.3 statistical error. What survives: μ_g IS the single largest systematic lever (mass group 0.38 vs rate group 0.22 km/s/Mpc per 1σ), so the obstruction becomes live when spectral-siren precision reaches ~ 1 km/s/Mpc (O5/XG) — then an independently pinned or z -dependent peak is mandatory. Conservative caveat: the restricted $H_0 \in [65, 77]$ prior compresses couplings (lower bounds), but the verdict does not hinge on it.

The nanohertz band: the GW background across pulsar-timing arrays (E68)

E68 cross-PTA GWB anatomy — pure degeneracy-sliding; γ marches with time span. First value/shape battery in the nanohertz band, on real validated chains: NANOGrav 15yr (HD-correlated), EPTA DR2full (HD), PPTA DR3 (CRN), + EPTA DR2new as robustness; CPTA has no public GWB chain. D1: all pairs mutually consistent (1.0–1.8 σ), Gaussian-faithful (sliced-W ratios 0.98–1.01). D2 HEADLINE: the constraint geometry is UNIVERSAL ($\rho(A, \gamma) = -0.85..-0.95$, $\psi = 110\text{--}116^\circ$ — one measurement kernel) and the pairwise mean offsets lie **94.6–99.9%** along the common A - γ degeneracy axis (mean 0.98; locked threshold 2/3) $\Rightarrow$ classified degeneracy-sliding: everything the PTAs disagree about lives along the direction a common power-law fit slides when read over different bands. D3: the SMBHB circular prediction $\gamma = 13/3$ is excluded $> 2\sigma$ by NANOGrav ALONE ($z = +3.03$; EPTA-full $+0.29$, PPTA $+1.25$) — the SH0ES pattern (E52): one experiment carries the narrative; no combined number fabricated (E58 lesson). POST-HOC (labelled): $\hat{\gamma}$ is STRICTLY MONOTONIC in observing span (10.3 $\rightarrow$ 24.7 yr: 2.79 $\rightarrow$ 3.25 $\rightarrow$ 3.87 $\rightarrow$ 4.21), including EPTA’s internal DR2new-vs-DR2full split (1.89 σ , same telescopes, offset also along the degeneracy) — the coherent signature of a spectrum that flattens at high frequency relative to a single power law (or a span-correlated systematic). Successor prediction on record: longer-span subsets of the same array should fit systematically steeper γ (preregisterable

at IPTA DR3 / NG 20yr). 4 contract tests; every headline independently re-derived; all probe values match the publications.

GW data beyond black holes: the neutron-star EOS transfer (E63)

E63 neutron-star mass-radius geometry. First transfer of the value/shape framework out of cosmology, into the neutron-star equation of state, in the (M, R) plane. Real posteriors: NICER pulse-profile for the low-mass star J0030 (Riley 2019 X-PSI ST+PST; Miller 2019 2-oval/3-oval) and the high-mass star J0740 (Riley 2021; Miller 2021), plus GW170817 tidal Λ mapped to (M, R) via the quasi-universal C–Love relation. Column identifications and every headline independently re-derived; 5 contract tests pass. METHODOLOGICAL LESSON: in (M, R) the axes are incommensurable ($M_\odot$ vs km), so the degeneracy ANGLE ψ is metric-dependent and meaningless (joint-whitening forces orthogonal long axes — a spurious ψ -gap= 90° on every pair until caught); the scale-free shape invariant is the correlation $\rho(M, R)$. RESULT D1: the NICER pipeline/model systematic is VALUE-dominated — every Riley-vs-Miller and spot-model difference is a central-value offset at fixed orientation (orientation-only value fraction 0.90–1.00, $\Delta\rho \leq 0.08$); the sharpest is J0740, a +1.75 km radius offset (Miller 14.05 vs Riley 12.30) plus a $3.2\times$ larger Miller ellipse. This is the OPPOSITE of the cosmology analog (the KiDS statistic-dependence, E48, was a precision/shape artifact). RESULT D3: $\rho(M, R)$ is organized by MASS REGIME and robust across independent pipelines — low-mass J0030 $\rho \approx +0.86$ (steep M–R banana) vs high-mass J0740 $\rho \approx +0.25$ (flat top of the M–R curve), regime $\Delta\rho = 0.61$, consistent across pipelines (0.84/0.89/0.86; 0.29/0.21) — the NS analog of “the kernel sets orientation” (organizer = location on the mass–radius curve). D4 (exploratory, flagged): X-ray J0030 vs GW170817 at $\sim 1.4 M_\odot$ is MIXED — both a ~ 1.5 km value offset (GW prefers smaller R , $\sim 1.1\sigma$) and $\rho: 0.86 \rightarrow 0.28$, but the GW ρ is C–Love-mapping-induced. D5: all six posteriors are near-pure ($\text{Tr } \rho^2 = 0.990\text{--}0.996$); the $\sigma^2/8$ Chernoff repackaging (E62) persists. Honest limits carried up front: non-Gaussian (multimodal) NICER marginals, the mapping dependence of the GW point, and the QI repackaging.

GWTC-5/O4b: the curved chirp-mass law reconfirmed out-of-sample (E71)

E71 GWTC-5/O4b curved-law repeat. Third disjoint catalog after GWTC-1/2.1/3 (train) and GWTC-4/O4a (E67). Prereg locked (commit d19a7a8) BEFORE any O4b file was opened; decision rules byte-identical to E67, only a declared structural file/group adapter added. GWTC-5.0 O4b PE (Zenodo 20276106 + 20348006): 104 per-event releases, 52.84 GB, ZERO dropped, fully disjoint (event dates 240413–250119, zero name-overlap with the 86-event O4a set, 12 events from the 2025 tail). D1 (the on-record E65/E67 prediction; elongated $\text{axr} \geq 3$, $n = 32$): median $|\psi_{\text{curve}} - \psi_{\text{meas}}| = 1.22^\circ < 2^\circ$ — PASS (E67 gave 1.26°). D2: curve beats tangent, 5.41° vs 8.23° . D3: the E40 tangent law replicates a third time (median 8.23° ; chirp-vs-total wins 88%; Spearman(err, axr) = -0.69 , the strongest of the three catalogs). Zero free parameters — each prediction uses only the event’s own median source M_c and q marginal. Waveform-robust: the curve tracks the measured axis WITHIN every waveform family to $< 0.7^\circ$; the per-event largest-analysis pick (5 of 104 events do carry a Mixed group; the locked rule uses it) shifts only the absolute orientation. Verification four legs: 17/17 contract tests; independent numpy re-derivation of D1/D2; one event hand-recomputed from raw HDF5 via a different linear-algebra path (match to 0.004°); computed medians reproduce the release’s published PESummaryTable masses (exact for most, to $< 0.6\%$ for a few late high-mass events), column identity confirmed catalog-wide. The mass-plane geometry law is now reproducible across THREE independent catalogs spanning the full GW history. POST-HOC (labelled): the largest curve residuals are all round posteriors ($\text{axr} \sim 1$), the ill-defined-orientation regime the axr gate excludes — not law-breaking physics; the E72 outlier atlas must control for axr before reading residuals as anomalies.

E72 GW geometry outlier atlas. Prereg locked (0d1cebfb) BEFORE any residual-axis correlation. Where does the constant- M_c curve law break? Searched the 32 elongated O4b events ($\text{axr} \geq 3$; residual $r = \text{err}_{\text{curve}}$ from the locked E71 results) against 9 pre-declared axes with Benjamini–Hochberg FDR ($q = 0.05$) and partial-Spearman controlling axr; Tukey outlier fence; E59 coherence lens. RESULT — a clean PHYSICAL NULL: none of precession (χ_p), aligned spin (χ_{eff}), mass ratio, redshift, chirp/total mass, or secondary mass (NSBH/mass-gap proxy) survives FDR — the constant- M_c geometry has NO source-class-dependent breakdown in O4b (strengthens claim 1). The only surviving correlate is waveform-model disagreement (a systematic; $\rho = -0.64$, $q_{\text{BH}} = 0.001$, robust to drop-one ≤ 0.036), and its NEGATIVE sign says not-physics: the events the curve fits best ($r \sim 0.01$ –

0.2°) are those where waveforms disagree most — the curve tracks the waveform-ensemble consensus better than any single waveform tracks another (measurement optics; the E59 lesson again). One marginal Tukey outlier (GW240507, 3.37° vs fence 3.23° ; low-SNR, asymmetric $q \sim 0.27$) logged as a WEAK preregisterable successor target, not a claim. Verify: 6 contract tests; the headline ρ and the Tukey fence independently re-derived (hand rank-Pearson; numpy percentile). Limits: $N = 32$ low power; loudness axis data-limited (6/32 picked groups carry a network-SNR column); eccentricity and higher-mode axes absent from the released PE (out of scope).